MIL-R-19445A (AER)
1 SEPTEMBER 1959
SUPERSEDING
MIL-R-19445(AER)
28 MARCH 1956

MILITARY SPECIFICATION

RACK, BOMB, AERO 65A SERIES

This specification has been approved by the
Bureau of Aeronautics, Department of the Navy.

1. SCOPE

1.1 Scope - This specification establishes the requirements for the manufacture, inspection and testing of one type of bomb rack to be known as Rack, Bomb, Aero 65A Series, and hereinafter referred to as rack.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on date of invitation for bids form a part of this specification:

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock Corner
PPP-C-843	Cushioning Material, Cellulosic
UU-P-271	Paper, Wrapping, Waterproofed Kraft
PPP-B-636	Boxes, Fiber

Military

JAN-P-100	Packaging and Packing for Overseas Shipment, General Specification
MIL-A-140	Adhesive, Water-Resistant, Waterproof Barrier Material
MIL-B-21034	Bomb Arming Unit Aero 2A, 2A-1, 2B
MIL-D-70327	Drawings, Engineering and Associated Lists
MIL-H-22162	Hook, Bomb Rack, Aero 65A Series
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-E-7894	Electric Power, Aircraft Characteristics of

FSC 1095

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MIL-P-116	Preservation, Methods of
MIL-T-7743	Testing; Store Suspension Equipment, General Specification for

STANDARDS

Federal

Fed. Test Method Std. 151	Metals, Test Methods

Military

MIL-STD-129	Marking for Shipment and Storage

PUBLICATIONS

Air Force - Navy Aeronautical Bulletin

ANA Bulletin No. 143	Specification and Standards; Use of

Consolidated Classification Committee

Consolidated Freight Classification Rules
(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois)

DRAWINGS

List of Drawings Index 55A44A69 Bomb Rack, Aero 65A Series and all drawings listed thereon

2.2 Availability of Documents

(1) Give both the title and number when requesting applicable documents. All requests should be made via the cognizant Inspector of Naval Material or the Bureau of Aeronautics Representative.

(2) Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, (CDS), 5801 Tabor Avenue, Philadelphia 20, Pennsylvania. Requests for drawings must contain a statement of justification for actual need in relation to a government contract.

2.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over all referenced documents.

(2) This Specification - This specification shall have precedence over all applicable subsidiary documents. Any deviation from this specification, or from subsidiary documents, where applicable, may be made only upon written approval

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(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials - Parts and materials used in the manufacture of the rack shall conform to the applicable specifications as specified herein or as listed on the applicable drawings. Material conforming to contractors' specifications may be used provided the specifications are approved by the procuring agency, are equivalent to the Government specifications and contain requirements for adequate tests of the material. The use of contractors' specifications shall not constitute a waiver of Government inspection. All materials shall be of the lightest practical weight and entirely suitable for the intended purpose. When materials are used which are procured under commercially recognized non-Government specifications, a certificate acceptable to the Government Inspector, showing compliance with such specifications, shall be furnished by the supplier of such material.

3.1.1 Critical and Strategic Materials - Materials shall be chosen on the basis of suitability and availability in this country in quantity and from consideration of the additional restrictions created during a national emergency. Critical and strategic materials shall be used only to the extent considered mandatory for satisfactory operation. The contemplated use of these materials shall be brought to the attention of the procuring agency with appropriate justification therefor and shall be subject to approval of the procuring agency.

3.1.2 Selection of Materials - Specification and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification shall be selected in accordance with ANA Bulletin No. 143.

3.1.2.1 Standard Parts - Standard parts (MS, AN or JAN) shall be used wherever they are suitable for the purpose, and shall be identified in the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used provided they possess suitable properties and are replaceable by the standard parts (MS, AN or JAN) without alteration and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.1.2.2 Weight Reduction - The contractor shall make every effort to reduce the equipment's over-all weight without impairing the function, operation or durability. All changes that are required to accomplish weight reduction shall be subject to written approval by the procuring agency prior to incorporation.

3.2 Design and Construction

3.2.1 Conformance to Drawings - No departure from or revision to the drawings specified in 2.1 shall be permitted without prior written approval by the procuring agency. In the event changes are approved (see 6.1), the contractor shall furnish the procuring agency with reproducible drawings covering the change. Conformance to the drawings will not relieve the contractor of the requirements of 3.4.

3.2.2 Input Power Requirements - This equipment shall perform satisfactorily under all applicable conditions specified in Specification MIL-T-7743, from a 28 volt power source with characteristics and limits as defined in Specification MIL-E-7894.

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3.2.3 Interchangeability - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-70327.

3.3 Performance - The performance of the rack shall be such that it conforms to all the requirements of this specification. The bomb rack shall be capable of suspending and releasing any store of a nominal weight between and including 100 and 1450 pounds and equipped with the suspension lugs spaced 14 inches center to center. The rack is designed for use in both dive, horizontal and other types of bombing. The rack may be actuated electrically or by the manual release system of the airplane and can be suspended or mounted, either externally as from an airplane wing or fuselage, or internally as within a bomb bay.

3.4 Workmanship - All details of workmanship shall be in accordance with high grade armament manufacturing practice for the quantity production of this type of equipment. All parts shall be free from excessive marks, grooving, indentations, cracks, flaws and any other injurious defects which may impair the serviceability of the assembled equipment. All electrical components and wiring shall be installed in accordance with the workmanship requirements of Specification MIL-E-5400. All workmanship shall be of a quality satisfactory to the procuring agency.

3.4.1 Manufacturing and Finish - Manufacturing processes and finishes shall be as shown on the applicable drawings specified in 2.1 unless otherwise approved by the procuring agency, (see 6.1). The responsibility shall be assumed by the contractor for any malfunction or failure of the parts or the assembled equipment to pass tests specified herein due to unsatisfactory fits, clearance, tolerances, internal stresses or finishes.

3.5 Correction of Defects - Parts not supplied by the Government shall be considered contractor's responsibility. Any defects in items, parts and materials discovered during the inspection and testing shall be corrected by the contractor at his expense. Such corrective measures as may be found necessary shall likewise be incorporated in all subsequent bomb racks offered for acceptance under the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests

(1) Preproduction Tests - Preproduction Tests are those tests conducted on racks representative of the equipments to be supplied under the contract, for the purpose of determining that the sample items meet all the requirements of the applicable specifications.

(2) Acceptance Tests - Acceptance Tests are those tests conducted on racks submitted for acceptance under the contract and shall consist of the following:

(a) Individual Tests - Individual Tests are those tests conducted on each rack submitted for acceptance under the contract.

(b) Sampling Tests - Sampling Tests are those tests conducted on a quantity of racks selected as specified herein and which have previously passed Individual Tests.

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4.2 Preproduction Tests - The preproduction samples shall consist of the first five racks fabricated by quantity production methods and tools. The preproduction samples shall be delivered for testing to a prescribed testing activity as directed by the procuring agency. Certification from the cognizant Government Inspector that the material used in the samples is in accordance with the requirements of this specification shall be forwarded with the samples. Approval of the preproduction samples shall be by the procuring agency upon satisfactory completion of all of the tests listed below. Any defects or failures occuring during these tests shall be cause for tentative rejection of all the preproduction samples. All rejected sample racks will be returned to the contractor for rework and correction of defects if this procedure is considered practical and feasible by the procuring agency, otherwise the contractor will be required to submit five additional samples for retests in lieu of the five reworked samples. If these samples fail any of the required tests, further retests shall be at the discretion of the procuring agency. The preproduction tests shall be divided into three parts as follows:

(1) Basic Tests

(2) Major Tests

(3) Critical Tests

The preproduction samples must successfully pass each of the applicable tests. The distribution of the tests among the samples is given in the following subparagraphs and also in Table I.

4.2.1 Basic Tests - The required Basic Tests are specified in 4.4.1. Every preproduction sample shall be subjected to these tests.

4.2.2 Major Tests - The required Major Tests are specified in 4.4.2.1. All of the Major Tests shall be performed on each set of preproduction samples, however the number of major tests to which a sample is subjected and the order of testing shall be at the discretion of the testing agency unless specified otherwise by the procuring agency. Table I is a recommended testing schedule for the five samples.

4.2.3 Critical Tests - The required critical tests are specified in 4.4.2.2. Letters A, B, C, D, E are arbitrary designations assigned to the samples by the testing agency. The distribution of the critical tests among the five samples shall conform to the following schedules:

Sample	Test
A	4.4.2.2 (a) Vibration
B	4.4.2.2 (b) Shock Test, Dynamic
C	4.4.2.2 (b) Shock Test, Static 4.4.2.2 (d) Destructive Load Test
D	4.4.2.2 (c) Life Test (Bomb Portion)
E	Critical Test to be selected by the testing agency

The above schedule is included in Table I. Selection of the critical tests to which any additional samples will be subjected shall be at the discretion of the testing agency, unless otherwise specified by the procuring agency.

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TABLE I

TEST SCHEDULE

RACK, BOMB, AERO 65A SERIES

Preproduction and Production Tests
which are to be performed on the
preproduction racks by the testing agency.

TYPE OF TEST	SPECIFICATION	APPL. PARA.	DESCRIPTION OF TESTS	SAMPLE IDENTIFICATION AND ORDER OF TESTS				
				A	B	C	D	E
BASIC	MIL-R-19445A	4.4.1.1	Examination of Product	1	1	1	1	1
		4.4.1.2	Operational Tests	2	2	2	2	2
		4.4.1.3	Arming Mechanism Tests	3	3	3	3	3
		4.4.1.4	Electrical Fuze Arming Tests	4	4	4	4	4
MAJOR	MIL-T-7743	4.2	Physical Characteristics	5	5	5	5	5
		4.3	Manual Operation					6
		4.4	Electrical Characteristics					10
		4.5	Lag Tests					7
		4.6	Performance under Icing Conditions					8
		4.9	Hi-G Release Test		6	6		
		4.11	Performance under Sand Conditions					9
		4.12	Salt Spray Test		7			
CRITICAL	MIL-T-7743	4.7	Vibration Tests	6				
		4.8.1	Shock Test, Dynamic		8			
		4.8.2	Shock Test, Static			7		
		4.10	Life Test				6	
		4.13	Destructive Load Test			8		

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4.3 Acceptance Tests - The contractor shall be responsible for conducting the Individual Tests specified in 4.3.1 on each rack manufactured. Necessary sampling tests will be conducted in accordance with 4.3.2. The racks shall be submitted for acceptance by production lots. A production lot shall consist of racks produced under essentially the same conditions from production tools and having previously passed the Individual Tests in 4.3.1. A production lot size shall not exceed 300 unless approved by the procuring agency. Acceptance of each production lot will be determined by the samples' successfully passing the sampling tests of 4.3.2. If any defects or failures occur during these tests, the manufacturer shall be required to eliminate the faults by reworking the production lot. The reworked production lot shall again pass the tests of 4.3.1, after which five samples shall be submitted for retests, the retests to be conducted in accordance with the tests of 4.3.2. After the first retest, further retests shall be at the discretion of the procuring agency. Inspection and approval of materials in the process of manufacture shall not be construed as insuring the acceptance or approval of the finished product.

4.3.1 Individual Tests - The contractor shall conduct on every rack the Basic Tests of 4.4.1. These tests are given in Table II.

TABLE II

TEST SCHEDULE

RACK, BOMB, AERC 65A SERIES

Individual Tests to be performed on every rack by the manufacturing agency.

SPECIFICATION	APPLICABLE PARAGRAPH	DESCRIPTION OF TESTS
MIL-R-19445A	4.4.1.1	Examination of Product
MIL-R-19445A	4.4.1.2	Operational Tests
MIL-R-19445A	4.4.1.3	Arming Mechanism Tests
MIL-R-19445A	4.4.1.4	Electrical Fuze Arming Tests

4.3.2 Sampling Tests - A random sampling of five racks shall be selected by a Government Inspector from each production lot and submitted for tests. The sampling tests are to be conducted by the testing agency and shall be divided into three parts as the preproduction test of 4.2. For acceptance of the lot, the sample must successfully pass each of the applicable tests. The distribution of tests among the samples is specified in the following sub-paragraphs and in Table I.

4.3.2.1 Basic Tests - The required Basic Tests are specified in 4.4.1. Every production sample shall be subjected to these tests.

4.3.2.2 Major Tests - The required Major Tests are specified in 4.4.2.1, except that the tests in 4.4.2.1 (e) (g) (h) may be deleted at the discretion of the procuring agency. All tests shall be performed on each set of production lot samples, but the number of tests to which a sample is subjected and the order of testing shall be at the discretion of the testing agency unless specified otherwise by the Bureau of Aeronautics. A recommended testing schedule for the five samples is included in Table I.

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4.3.2.3 Critical Tests - The required Critical Tests are specified in 4.4.2.2. The distribution of the Critical Tests among the five samples must follow the schedule of 4.2.3.

4.4 Test Methods

4.4.1 Basic Tests

4.4.1.1 Examination of Product - Each rack shall be inspected to insure that:

(1) The rack has been properly assembled and adjusted.

(2) The electrical circuits and connections are correct, positive, and secure.

(3) The dimensions are within tolerance.

(4) The proper hardness levels are maintained.

(5) The proper plating is maintained.

(6) The rack operates manually and electrically.

(7) The finish and workmanship are in accordance with this specification and all the applicable specifications and drawings listed herein.

4.4.1.2 Operational Tests - Each completely assembled rack shall be mounted in a horizontal attitude on a test stand which closely simulates the mounting of the rack on an aircraft. The rack shall be latched (unloaded) and a safetying pin inserted through the safetying pin hole in the rack frame. The rack shall be released manually and then electrically to insure that the hooks remain latched. Upon removal of the safetying pin, the hooks shall fully open. The rack shall be relatched and without inserting the safetying pin, the manual release shall be actuated to fully open the hooks. This operation shall be repeated three times and the travel and pull of the release cable for "release" and the travel and pull of the cable for "full stroke" shall be measured and shall conform to the "Rack Release Data" table on Bureau of Aeronautics Drawing 55A44R81. The rack shall then be loaded with a simulated 1000 pound store and the release circuit energized electrically with a 28 volt, 0.020 second pulse to release the store clear of the rack. This operation shall be repeated three times.

4.4.1.3 Arming Mechanism - A bomb arming wire, Mk 1 (AN-M6A2), shall be inserted in each of the arming mechanisms and the arming solenoids individually energized. Apply a proof load of 100 pounds horizontally forward and vertically downward on the nose arming mechanism and horizontally aft and vertically downward on the tail arming mechanism. Each load shall be applied separately through the arming wire. The arming wire swivel loop shall not pull out of the retainer nor shall any damage to the item occur. With the arming solenoid de-energized, the force required to withdraw the arming wire shall not be less than 2 pounds nor more than 8 pounds for any direction of pull in the forward quadrant for the nose arming mechanism and in the aft quadrant for the tail arming mechanism.

4.4.1.4 Electrical Fuze Arming Mechanism (when applicable)

(1) Cable Tension Test - A pull of one to three pounds exerted on the arming plug shall not extend the plug and cable from their fully retracted position. A pull of 15 ± 2.5 pounds shall extend the plug and cable fully. Upon the removal of this force, the plug and cable shall return to the fully retracted position.

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(2) Operational Test - A minimum of five bomb releases will be made with the applicable arming unit described in Specification MIL-B-21034 connected to the bomb. The arming unit circuit shall be monitored in order to determine whether the prescribed electrical arming impulse is transmitted to the bomb.

(3) Electrical Continuity Checks - The necessary electrical continuity checks will be made to see that the arming unit circuit, in each phase of its operational cycle, conforms to the electrical circuit and operation sequence of Figure 1.

4.4.2 Tests of Specification MIL-T-7743 - The following tests of MIL-T-7743 shall be performed with procedures as outlined therein.

4.4.2.1 Major Tests

(a) 4.2 Physical Characteristics

(b) 4.3 Manual Operation

(c) 4.4 Electrical Characteristics

(d) 4.5 Lag Tests

(e) 4.6 Performance under Icing Conditions

(f) 4.9 Hi-G Release Test

(g) 4.11 Performance under Sand Conditions

(h) 4.12 Salt Spray Test

4.4.2.2 Critical Tests

(a) 4.7 Vibration Test

(b) 4.8 Shock Test (Loads and modifications given in section 4.5 of this specification)

(c) 4.10 Life Test

(d) 4.13 Destructive Load Test (Static yield tests)

4.5 Bomb Rack Test Load

4.5.1 Static Test Loads - The following are the critical limit inertia loads on the rack when carrying the 1000 pound, low-drag G. P. Bomb, Mark 83. These loads shall be used for the tests where applicable.

Condition	V#	S#	D#	R. M. "#	Y. M. "#
Flight	-4970	4320	-3530	-74200	82500
Catapulting	-3000	±1500	-9000	±16800	±5600
Arrested Landing	-3000	±1500	9000	±16800	±8400

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NOTES (1) Sign Convention for Static Test Loads

V - up loads are positive

S - loads to the right are positive when looking aft

D - forward loads are positive

P. M. - nose up pitching moments are positive

Y. M. - nose right yawing moments are positive when looking aft

(2) Design yield loads are 1.15 and ultimate loads are 1.5 times the limit values in the table.

(3) All loads and moments are at the center of gravity of the store.

(4) Sway braces are located 10 inches forward and 10 inches aft of the center of gravity at an angle of 30° on each side of the vertical center line of the store.

4.5.2 Dynamic Shock Test Loads - The loads specified for the Shock Test are the loads on the rack when carrying a 1000 pound, low-drag G. P. bomb (Mark 83). The center of gravity of the bomb is approximately 8 inches below the bearing surfaces of the lugs and 7.0 inches aft of the forward lug. The dynamic tests specified in MIL-T-7743 shall be modified where necessary so that the tests are accomplished in each of the six directions parallel to the three principle axes. The load factors (G Values) shall be as follows:

Direction	G Values
Vertical Down	11.0
Vertical Up	5.5
Longitudinal Forward	9.9
Port	7.2
Starboard	7.2

Notes (3) and (4) of 4.5.1 apply to the dynamic portion.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Unit Packaging

5.1.1 Level A

5.1.1.1 Preservation - Each rack shall be preserved in accordance with the requirements of method IA of Specification MIL-P-116, without the use of preservative compound.

5.1.1.2 Unit Packaging - Unless otherwise specified, cushioning shall be provided within the unit packaged to prevent rupture of the method IA barrier, using materials set forth in Specification PPP-C-843 and JAN-P-100. Unit containers shall be fiberboard boxes conforming to Specification PPP-B-636. Each container shall contain one complete rack (preserved) which shall be snugly packed.

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5.1.2 Level C

5.1.2.1 Preservation and packing shall be in conformance with commercial standards.

5.2 Packing - Unless otherwise specified in the contract or order, all racks shall be packed for Level B and C shipment. Shipping containers shall contain the same number of unit packages, shall be uniform in size, and snugly packed. The gross weight, when packed for shipment, shall not exceed 200 pounds.

5.2.1 Packing Level B and Level C - The unit packages shall be placed in substantial exterior shipping containers so constructed as to insure acceptance by common or other carrier for safe transportation at lowest rates to the point of delivery specified in the contract or order. Containers shall conform to the requirements of the Consolidated Freight Classification Rules in effect at the time of shipment, except that fiberboard, when used, shall have a minimum Mullen Test of 200 pounds. Containers shall be capable of withstanding storage, rehandling, and reshipment without the necessity of repacking.

5.2.2 Packing, Level A - When specified in the contract or order, the unit packages shall be packed in cleated plywood, or nailed wood boxes conforming to Specification PPP-B-601 (overseas type) or PPP-B-621 (overseas class). Boxes shall be furnished with a sealed case liner fabricated from case liner material conforming to Specification UU-P-271 and sealed with adhesive conforming to Specification MIL-A-140.

5.3 Marking and Labeling

5.3.1 Unit Packages - Each unit package shall be durably and legible marked or labeled with the following information and in accordance with the requirements of MIL-STD-129.

Bomb Rack Aero 65A (Series Letter, e.g. A-1, A-2, etc.)

Military Specification No.

Preserved (Date)

Domestic or Export Packed (as applicable)

Government Order No. (or Contract No. if Order No. is not assigned)

Names of Contractors (and name of manufacturer, if not the same)

5.3.2 Shipping Containers - All shipping containers shall be marked in accordance with applicable requirements of MIL-STD-129.

6. GENERAL NOTES

6.1 Compliance with Drawings and Specifications - Conflict Elimination - Insofar as the contractor is required by the provisions of this specification to comply strictly to the drawings listed in 2.1, which may make it impossible in some instances also to comply with the performance and test requirements specified herein, the contractor in that event should promptly notify the procuring agency of such fact, and appropriate changes will be made in the applicable drawings, parts or performance requirements as authorized by the procuring agency.

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6.2 Availability of Dummy Stores for Testing - The contractor shall submit requests to the procuring agency for dummy and inert stores to be used for testing as required by this specification.

6.3 Information Required With Bids - Each bid shall be accompanied by a statement as to whether or not any exceptions are being taken to the requirements shown in the drawings and specifications. If exceptions are being taken or deviations are anticipated, alternative proposals shall be indicated in the statement and submitted therewith.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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